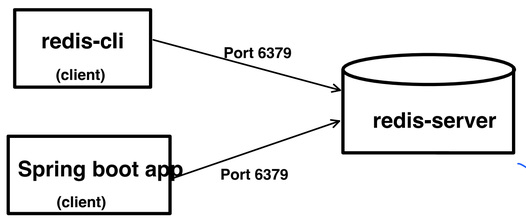
3. Installation and Running Redis

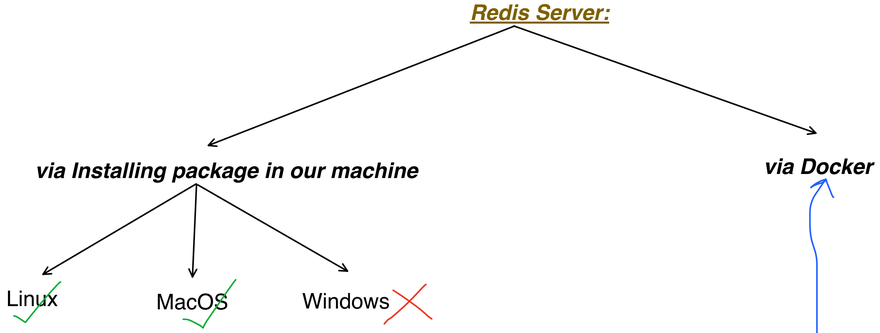
Thursday, 8 October 2026 2:07 PM

What we want to setup:



- Redis is a separate server.
- ***redis-cli*** is just a ready-made client. Our Spring Boot app will be another client.
- Redis listens on port ***6379*** by default.

Redis Server:



- In Redis Architecture part, we have seen that Redis internally make use of Linux feature epoll (similar to kqueue in MacOS)
- But native Windows have limited support for Redis. So to run Redis on Windows, we can use

Which one we will follow to start the Redis Server then?

- We will do the set up of Redis server via both Installing package in MacOS (similar to Linux) and via Docker.
- And in future topics of Redis, we will run the Redis server via Docker only, so that we all are on same page.

Install Redis on Mac

Step 1: Check that Homebrew is installed

```
brew --version
```

```
shrayanshjain@Shrayanshs-MacBook-Pro-2 ~ % brew --version
Homebrew 6.0.6
shrayanshjain@Shrayanshs-MacBook-Pro-2 ~ %
```

Step 2: Install Redis

```
brew install redis
```

Homebrew downloads Redis and installs both redis-server and redis-cli.

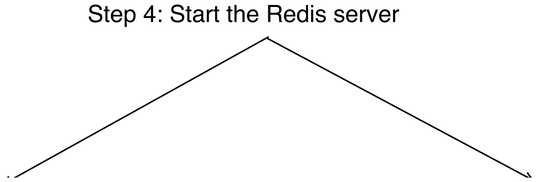
```
shrayanshjain@Shrayanshs-MacBook-Pro-2 ~ % brew install redis
==> Auto-updating Homebrew...
Adjust how often this is run with `$HOMEBREW_AUTO_UPDATE_SECS` or disable with
`$HOMEBREW_NO_AUTO_UPDATE=1`. Hide these hints with `$HOMEBREW_NO_ENV_HINTS=1` (see `man brew`).
==> Downloading https://ghcr.io/v2/homebrew/core/portable-ruby/blobs/sha256:e0088dff5614b39387300136ec7a5f95bf1e07589547245c919524fc9e8b4197
######################################################################################################################################## 100.0%
==> Pouring portable-ruby-4.0.7.arm64_big_sur.bottle.tar.gz
```

Step 3: Validate

```
redis-server --version
redis-cli --version
```

```
shrayanshjain@Shrayanshs-MacBook-Pro-2 ~ % redis-server --version
Redis server v=8.10.2 sha=00000000:1 malloc=libc bits=64 build=764b3e45065a66fb
shrayanshjain@Shrayanshs-MacBook-Pro-2 ~ % redis-cli --version
redis-cli 8.10.2
shrayanshjain@Shrayanshs-MacBook-Pro-2 ~ %
```

Step 4: Start the Redis server



Way 1: Run it as a background service

```
brew services start redis     # start
brew services stop redis      # stop
```

Why use it: Redis keeps running when we close the terminal.
Use this when we are building apps.

Way 2: Run it in the foreground

```
redis-server
```

Why use it: you see the logs live, and Ctrl-C stops it. Use this when you are learning or debugging.

```
shrayanshjain@Shrayanshs-MacBook-Pro-2 ~ % redis-server
88034:C 08 Oct 2026 17:28:17.769 * oO0OoO0OoO0Oo Redis is starting oO0OoO0OoO0Oo
88034:C 08 Oct 2026 17:28:17.769 * Redis version=8.10.2, bits=64, commit=00000000, modified=1, pid=88034, just started
88034:C 08 Oct 2026 17:28:17.769 # Warning: no config file specified, using the default config. In order to specify a config file use redis-ser
ver /path/to/redis.conf
88034:M 08 Oct 2026 17:28:17.771 * Increased maximum number of open files to 10032 (it was originally set to 256).
88034:M 08 Oct 2026 17:28:17.771 * monotonic clock: ARM CNTVCT @ 1000 ticks/us
                _._
           _.-``__ ''-._                Redis Open Source
      _.-``    `.  `_.  ''-._          8.10.2 (00000000/1) 64 bit
  .-`` .-```.  ```\/    _.,_ ''-._     Running in standalone mode
 (    '      ,       .-`  | `,    )     Port: 6379
 |`-._`-...-` __...-.``-._|'` _.-'|     PID: 88034
 |    `-._   `._    /     _.-'    |
 `-._    `-._  `-./  _.-'    _.-'
 |`-._`-._    `-.__.-'    _.-'_.-'|
 |    `-._`-._        _.-'_.-'    |           https://redis.io
  `-._    `-._`-.__.-'_.-'    _.-'
 |`-._`-._    `-.__.-'    _.-'_.-'|
 |    `-._`-._        _.-'_.-'    |
  `-._    `-._`-.__.-'_.-'    _.-'
      `-._    `-.__.-'    _.-'
          `-._        _.-'
              `-.__.-'

88034:M 08 Oct 2026 17:28:17.772 # WARNING: The TCP backlog setting of 511 cannot be enforced because kern.ipc.somaxconn is set to the lower va
lue of 128.
88034:M 08 Oct 2026 17:28:17.773 * Server initialized
88034:M 08 Oct 2026 17:28:17.773 * Ready to accept connections tcp
```

Connect with redis-cli

```
redis-cli
```

Its a health check, if Redis is alive, it will reply PONG

```
127.0.0.1:6379> PING
PONG
127.0.0.1:6379> SET greeting "hello redis"
OK
127.0.0.1:6379> GET greeting
"hello redis"
127.0.0.1:6379> exit
```

Just storing value under the key "greeting" and then reading it back.

```
shrayanshjain@Shrayanshs-MacBook-Pro-2 ~ % redis-cli

127.0.0.1:6379> PING
PONG
127.0.0.1:6379> SET greeting "hello redis"
OK
127.0.0.1:6379> GET greeting
"hello redis"
127.0.0.1:6379> exit
shrayanshjain@Shrayanshs-MacBook-Pro-2 ~ %
```

Check the configurations of our running redis-server

```
redis-cli CONFIG GET '*'
```

This will return all the configurations. For specific config, we can do

```
redis-cli CONFIG GET port
```

```
redis-cli CONFIG GET maxmemory
```

Now question is, I don't need this default configurations, I need control over this.

Then we can write our own configuration and while starting the redis server, we can tell which configuration to use.

Lets create the file ~/redis-lab/my-redis.conf

```
shrayanshjain@Shrayanshs-MacBook-Pro-2 redis-lab % mkdir -p ~/redis-lab/
shrayanshjain@Shrayanshs-MacBook-Pro-2 redis-lab % cd ~/redis-lab
shrayanshjain@Shrayanshs-MacBook-Pro-2 redis-lab % pwd
/Users/shrayanshjain/redis-lab
shrayanshjain@Shrayanshs-MacBook-Pro-2 redis-lab %
```

Lets say we are adding below config in the "my-redis.conf" file

port 6380
bind 127.0.0.1
dir /Users/shrayanshjain/redis-lab/data
loglevel notice
maxmemory 256mb

Line	Meaning
port 6380	Listen on 6380
bind 127.0.0.1	Accept connections only from this Mac
dir ...	The folder where Redis saves its data files
loglevel notice	How much detail the log shows
maxmemory 256mb	Redis may use at most 256 MB of RAM for data

Lets start the server now:

redis-server ~/redis-lab/my-redis.conf

```
shrayanshjain@Shrayanshs-MacBook-Pro-2 ~ % redis-server ~/redis-lab/my-redis.conf

*** FATAL CONFIG FILE ERROR (Redis 8.10.2) ***
Reading the configuration file, at line 3
>>> 'dir /Users/shrayanshjain/redis-lab/data'
No such file or directory
shrayanshjain@Shrayanshs-MacBook-Pro-2 ~ % redis-server ~/redis-lab/my-redis.conf
89855:C 09 Oct 2026 14:24:29.315 * oO0OoO0OoO0Oo Redis is starting oO0OoO0OoO0Oo
89855:C 09 Oct 2026 14:24:29.315 * Redis version=8.10.2, bits=64, commit=00000000, modified=1, pid=89855, just started
89855:C 09 Oct 2026 14:24:29.315 * Configuration loaded
89855:M 09 Oct 2026 14:24:29.315 * Increased maximum number of open files to 10032 (it was originally set to 256).
89855:M 09 Oct 2026 14:24:29.315 * monotonic clock: ARM CNTVCT @ 1000 ticks/us
                _._
           _.-``__ ''-._
      _.-``    `.  `_.  ''-._           Redis Open Source
  .-`` .-```.  ```\/    _.,_ ''-._      8.10.2 (00000000/1) 64 bit
 (    '      ,       .-`  | `,    )     Running in standalone mode
 |`-._`-...-` __...-.``-._|'` _.-'|     Port: 6380
 |    `-._   `._    /     _.-'    |     PID: 89855
  `-._    `-._  `-./  _.-'    _.-'
 |`-._`-._    `-.__.-'    _.-'_.-'|
 |    `-._`-._        _.-'_.-'    |           https://redis.io
  `-._    `-._`-.__.-'_.-'    _.-'
 |`-._`-._    `-.__.-'    _.-'_.-'|
 |    `-._`-._        _.-'_.-'    |
  `-._    `-._`-.__.-'_.-'    _.-'
      `-._    `-.__.-'    _.-'
          `-._        _.-'
              `-.__.-'

89855:M 09 Oct 2026 14:24:29.315 # WARNING: The TCP backlog setting of 511 cannot be enforced because kern.ipc.somaxconn is set to the lower
lue of 128.
89855:M 09 Oct 2026 14:24:29.316 * Server initialized
```

Connect with redis-cli:

redis-cli -h 127.0.0.1 -p 6380 PING

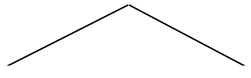
```
shrayanshjain@Shrayanshs-MacBook-Pro-2 ~ % redis-cli -h 127.0.0.1 -p 6380 PING

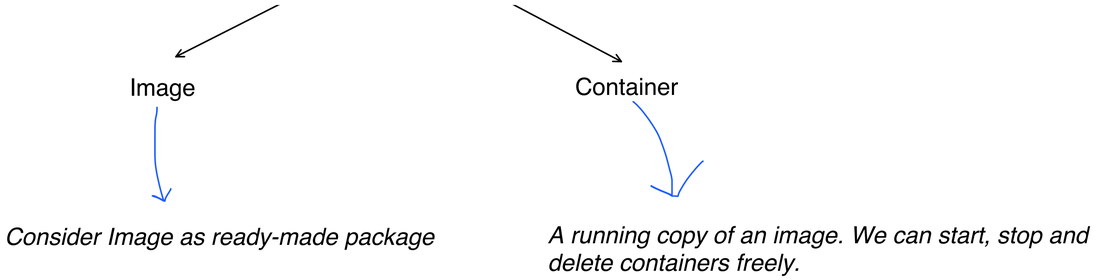
PONG
shrayanshjain@Shrayanshs-MacBook-Pro-2 ~ %
```

Run Redis with Docker

Please Note: Docker we will cover in-depth as part of separate playlist.

But for now, understand 2 terms:





- Without Docker: Install Redis on our Mac, configure it, and start the server.
- With Docker: Download the Redis image and start a container from it.

Step 1: Install Docker

Install Docker Desktop from Docker's official site (Redis's Docker page links to the guide for each OS).

https://redis.io/docs/latest/operate/oss_and_stack/install/install-stack/docker/

Once docker installed, then check

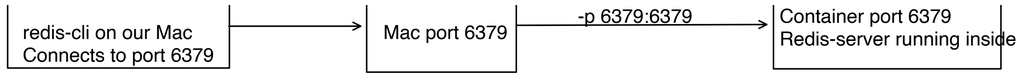
docker --version

```
shrayanshjain@Shrayanshs-MacBook-Pro-2 ~ % docker --version
Docker version 28.5.2, build ecc6942
shrayanshjain@Shrayanshs-MacBook-Pro-2 ~ %
```

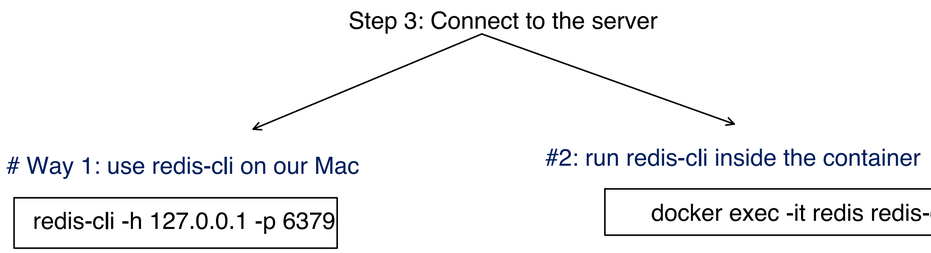
Step 2: Run Redis

```
docker run -d --name redis -p 6379:6379 redis
```

Line	Meaning
docker run	Creates and start a new container
-d	Run in the background
--name redis	Name the container 'redis', so that we can refer it
-p 6379:6379	Port mapping: Mac port : container port
redis	Image to use. Docker download it first time




```
shrayanshjain@Shrayanshs-MacBook-Pro-2 redis-lab % docker run -d --name redis -p 6379:6379 redis
Unable to find image 'redis:latest' locally
latest: Pulling from library/redis
a393f1d59f88: Pull complete
3adec69eeaf2: Pull complete
4f4fb700ef54: Pull complete
bbeda6b4abb7: Pull complete
cbce0b02cac7: Pull complete
7d58206b798c: Pull complete
73c2f9166d69: Pull complete
Digest: sha256:2c2dff791878316e3b083188be0d578423b24813cdda5de0e80ea4f7b3e6bfd6
Status: Downloaded newer image for redis:latest
b5456b6fe7610f8498fa855c8951b9dd717358a1bb8998677b477e64d9bb526f
shrayanshjain@Shrayanshs-MacBook-Pro-2 redis-lab %
```



docker ***exec*** runs a command inside a running container, so that we don't have to install redis-cli separately on our machine.

docker exec	Execute a command inside an existing, running container
-it	Lets us type commands
redis	Name of the running container
redis-cli	Command to execute inside that container

```
shrayanshjain@Shrayanshs-MacBook-Pro-2 redis-lab % docker exec -it redis redis-cli
127.0.0.1:6379> ping
PONG
127.0.0.1:6379>
```

Connect from Springboot

Step 1: Add the dependency

```
<dependency>
   <groupId>org.springframework.boot</groupId>
   <artifactId>spring-boot-starter-data-redis</artifactId>
</dependency>
```

This brings "Lettuce" the Redis client library that spring uses by-default.

Step 2: Tell the app, where the redis is

application.properties

```
spring.data.redis.host=localhost
spring.data.redis.port=6379
```

That’s all we need, lets write sample connect and validate the connection.

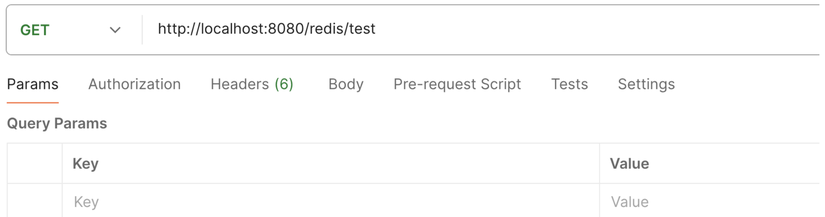
```
@RestController
public class RedisTestController{

@Autowired
private String RedisTemplate redis;

@GetMapping("/redis/test")
public String test() {
   redis.opsForValue().set("greeting", "hello from springboot");
   return redis.opsForValue().get("greeting");
  }
}
```

Springboot automatically set up the connection for us.

Invoke the api:



Body Cookies Headers (5) Test Results Status: 200 OK T

Pretty Raw Preview Visualize Text

```
1  hello from spring boot
```

Also double check from redis-cli:

```
shrayanshjain@Shrayanshs-MacBook-Pro-2 redis-lab % docker exec -it redis redis-cli GET greeting
"hello from spring boot"
shrayanshjain@Shrayanshs-MacBook-Pro-2 redis-lab %
```